ModelIt!

Teacher's Guide

How Do We Build a Sustainable City?

Modeling Urban Systems, Energy Networks, and Resource Flows for Resilient Communities

Quick Reference

Grade Level	Grade 12L3
NGSS Standard	HS-ESS3-3
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-ESS3-3

Create a computational simulation to illustrate the relationships among management of natural resources, the sustainability of human populations, and biodiversity.

Three-Dimensional Learning

Science Practice	**Using Mathematics and Computational Thinking** — Students use computational simulations to model the relationships among urban resource management, sustainability metrics, and population dynamics, quantifying the impact of different intervention strategies.
Disciplinary Core Idea	**ESS3.C Human Impacts on Earth Systems** — Students model how urban design decisions affect energy consumption, carbon emissions, water cycles, waste accumulation, and ecosystem health, analyzing the cumulative impact of cities on Earth systems.
Crosscutting Concept	**Systems and System Models** — Students analyze how the sustainability of a city emerges from the interactions among energy, transportation, water, waste, and land use systems, demonstrating that optimizing individual systems does not optimize the whole.

CAST Testing Alignment

MULTIPLE CHOICE:

A city achieves 80% renewable energy but maintains 90% car mode share and 30% waste diversion. Based on the urban systems model, what is the most likely impact on the overall Urban Sustainability Index and why?

CONSTRUCTED RESPONSE:

Using evidence from your model, explain why a city that simultaneously implements moderate improvements across all urban systems (energy, transit, waste, water, green space) achieves a higher Urban Sustainability Index than a city that maximizes improvement in one system while neglecting others. Reference the concepts of system interdependence and feedback loops.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Cities are the defining habitat of the 21st century. More than 4.4 billion people — 56% of humanity — live in urban areas, and that number is projected to reach 6.7 billion by 2050. Cities occupy only 3% of Earth's land surface but consume 75% of natural resources, produce 70% of global carbon emissions, and generate 50% of global waste. How we design, power, and manage cities will largely determine whether humanity can address climate change and build a sustainable future.

Urban Metabolism: The City as Ecosystem

The concept of urban metabolism treats cities as ecosystems with inputs (energy, water, food, materials), internal processes (manufacturing, transportation, building operations), and outputs (emissions, wastewater, solid waste, heat). A linear urban metabolism takes resources, uses them once, and discards the waste. A circular urban metabolism closes loops: waste becomes feedstock, wastewater becomes irrigation, waste heat warms buildings, and organic waste becomes compost for urban farms. The most sustainable cities are those that maximize the circularity of their metabolism — reducing inputs, extending the useful life of materials, and recovering value from outputs.

Transportation and Land Use

Transportation is the single largest source of emissions in most American cities, and it is inseparable from land use. Sprawling cities with separated residential, commercial, and industrial zones force long car commutes and make public transit economically unviable. Dense, mixed-use cities with transit-oriented development enable walking, cycling, and public transit as primary modes. The car-dependent model consumes enormous amounts of land for roads and parking — in many US cities, 30-50% of urban land is dedicated to vehicles. Reclaiming this land for housing, parks, and productive use is one of the largest untapped sustainability opportunities.

Green Infrastructure and Ecosystem Services

Traditional engineering approaches to urban challenges — concrete stormwater drains, air conditioning, water treatment plants — are energy-intensive and expensive. Green infrastructure provides many of the same services using natural processes at lower cost: urban forests absorb CO2, filter air pollution, and reduce temperatures by 2-5 degrees through evapotranspiration; bioswales and permeable surfaces manage stormwater naturally, reducing flooding and sewer overflows; green roofs insulate buildings, reducing heating and cooling energy by 25-40%. These ecosystem services compound as coverage increases, creating a tipping point where green infrastructure significantly reduces the need for engineered solutions.

Equity in Sustainability

Environmental sustainability and social equity are inseparable. Low-income neighborhoods and communities of color consistently bear disproportionate environmental burdens: more air pollution from highways and industrial facilities, less green space, higher urban heat island temperatures, more flooding from inadequate infrastructure, and less access to public transit. Sustainability interventions that ignore equity can worsen disparities — for example, green gentrification occurs when parks and bike lanes increase property values, displacing long-term residents who most need the improvements. Equitable sustainability requires that the communities most affected by environmental harm receive the benefits of sustainability investments first.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify ten key components of the urban sustainability system: population density, solar availability, water supply, energy mix, transportation mode share, water recycling, waste diversion, green infrastructure, carbon emissions, and urban sustainability index.

E **Establish Relationships**

Students map relationships showing how energy mix and transportation directly drive emissions, how green infrastructure provides non-linear ecosystem service benefits, and how all systems interact to produce the composite sustainability index.

V **Visualize & Model**

Students build a ten-component computational model showing the interconnected urban metabolism with feedback loops between emissions, heat island effects, energy demand, and green infrastructure benefits.

E **Evaluate Outcomes**

Students run business-as-usual, energy-only, and integrated redesign scenarios to quantify why systemic change produces dramatically better outcomes than single-system optimization.

R **Revise & Extend**

Students add food system localization, social equity index, or climate resilience score to model the full complexity of sustainable urban development including justice and adaptation dimensions.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"By 2050, two out of every three humans will live in a city. Cities consume 75% of global energy and produce 70% of carbon emissions. The way we build cities is the way we build our future. Today you are going to redesign what a city can be."*

Watch For

Show cover image. Ask students: what is the most sustainable city you can think of? What makes it different from most American cities? Build a class list of sustainability features.

Slide 2: Learning Objectives

3 min

Say: *"Today you are building a model of an entire city as a system. You will discover why no single solution — not even 100% renewable energy — can make a city sustainable, and why systemic redesign is the only path forward."*

Watch For

Have students read objectives. Pre-teach 'urban metabolism' by comparing a city to a biological organism: inputs, processes, outputs, waste. Ask: what does a healthy metabolism look like?

Slide 3: Big Question

3 min

Say: *"Can we redesign cities as self-sustaining ecosystems? The answer depends on whether we are willing to change not just our energy sources but our transportation, our waste systems, our water management, and our relationship to green space — all at once."*

Watch For

Quick-write: Students describe their ideal sustainable city in 3-5 sentences. What features would it have? How would daily life be different? Share out and catalog ideas.

Slide 4: LEVER Framework

2 min

Say: *"We will Label ten components of the urban system, Establish how they connect and create feedback loops, Visualize the full urban metabolism model, Evaluate three very different city designs, and Refine by adding equity and resilience dimensions."*

Watch For

Preview each LEVER step. Emphasize that cities are the most complex human-made systems on Earth — modeling even ten components reveals surprising insights about how they work.

Slide 5: Activity 1: Components

8 min

Say: *"Three of these components are determined by geography and demographics — the constraints every city faces. Seven are design choices that urban planners, engineers, and communities can change. Sort them and discuss: which design choices have the most impact?"*

Watch For

Students sort external versus internal components. Debate whether Population Density is truly external or partially a design choice through zoning policy. Use this to discuss how policy converts external constraints into internal variables.

Slide 6: Activity 2: Connections

8 min

Say: *"Energy mix drives emissions. Transportation mode share affects land use and air quality. Green infrastructure reduces heat islands and stormwater costs. But these all interact — and some interactions create reinforcing feedback loops that either accelerate sustainability or lock in unsustainability."*

Watch For

Students draw arrows showing positive and negative relationships. Identify the heat island feedback loop: more heat -> more cooling energy -> more emissions -> more heat. Identify how green infrastructure breaks this loop.

Slide 7: Activity 3: Simulation

10 min

Say: *"Run three city configurations: a typical American city, the same city with 80% renewable energy, and the same city with integrated sustainable redesign across all systems. The difference between scenarios two and three will surprise you."*

Watch For

Students run all three scenarios and compare results. Key discussion: why does the energy-only scenario achieve only modest sustainability improvement? What does this tell us about systemic change?

Slide 8: Discoveries

5 min

Say: *"The model reveals that no single solution works alone. A city with 100% renewable energy but car-dependent sprawl is still unsustainable. Sustainability is a systems property that emerges from simultaneous improvements across energy, transit, waste, water, and green space."*

Watch For

Class discussion of discoveries. Ask: which finding has the biggest implications for how cities should invest their sustainability budgets? Why is this a systems problem, not an energy problem?

Slide 9: STEM Challenge

12 min

Say: *"The mayor has hired your team. Population 500,000. Current sustainability: poor. Budget: limited. Mandate: net-zero by 2050, equitably. Your model data tells you what works. Now design the plan."*

Watch For

Groups design master plans with quantitative targets, implementation timelines, equity provisions, and model-based justifications. Present and compare: which plan achieves the best sustainability score? Which is most equitable? Which is most realistic?

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Population Density, Solar Radiation Availability, Regional Water Supply
- **INTERNAL:** Energy Mix Composition, Transportation Mode Share, Water Recycling Rate, Waste Diversion Rate, Green Infrastructure Coverage, Carbon Emissions Rate, Urban Sustainability Index

Relationships

- **Population Density to Carbon Emissions Rate:** NEGATIVE (-) — Higher population density enables more efficient shared infrastructure — public transit, district heating, compact buildings — reducing per-capita emissions. Sprawling low-density development forces car dependency and energy-intensive single-family housing.
- **Energy Mix Composition to Carbon Emissions Rate:** NEGATIVE (-) — A higher proportion of renewable energy directly reduces carbon emissions per unit of energy consumed. Shifting from fossil fuels to solar, wind, and geothermal is the most direct pathway to emissions reduction.
- **Transportation Mode Share to Carbon Emissions Rate:** NEGATIVE (-) — Shifting from private vehicles to walking, cycling, and public transit reduces per-capita transportation emissions by 50-90%. Transportation is typically the largest emissions sector in American cities.
- **Green Infrastructure Coverage to Carbon Emissions Rate:** NEGATIVE (-) — Green infrastructure reduces emissions both directly (trees sequester carbon) and indirectly (evapotranspiration reduces urban heat, which reduces cooling energy demand, which reduces power plant emissions).
- **Waste Diversion Rate to Carbon Emissions Rate:** NEGATIVE (-) — Diverting organic waste from landfills prevents methane production (a greenhouse gas 80x more potent than CO2 over 20 years). Recycling reduces the energy needed to produce new materials from raw resources.
- **Water Recycling Rate to Urban Sustainability Index:** POSITIVE (+) — Higher water recycling reduces demand on regional water supply, decreases energy used for water treatment and pumping, and reduces wastewater discharge that degrades waterways — contributing to multiple sustainability dimensions simultaneously.
- **Solar Radiation Availability to Energy Mix Composition:** POSITIVE (+) — Cities with greater solar resources can achieve higher renewable energy fractions more cost-effectively, making the transition from fossil fuels economically favorable without subsidies.
- **Green Infrastructure Coverage to Urban Sustainability Index:** POSITIVE (+) — Green infrastructure provides compounding ecosystem services — stormwater management, air purification, temperature regulation, biodiversity habitat, mental health benefits — that improve sustainability across environmental, social, and economic dimensions simultaneously.

Key Discoveries

- No single system change can make a city sustainable — the model reveals that renewable energy alone, transit improvements alone, or waste reduction alone each have limited impact because emissions and resource consumption are distributed across multiple interconnected systems
- Transportation mode share has a surprisingly large impact on overall sustainability because it affects not just emissions but land use (parking lots versus parks), public health (air quality, physical activity), equity (car dependency excludes low-income residents), and energy demand simultaneously
- Green infrastructure provides non-linear benefits — beyond a threshold of approximately 20-25% coverage, ecosystem services like stormwater management, temperature reduction, and air purification compound to significantly reduce the cost and energy demand of engineered infrastructure solutions

- The urban heat island effect creates a positive feedback loop: higher temperatures increase cooling energy demand, which increases emissions from fossil fuel power plants, which contributes to warming — green infrastructure breaks this loop by providing passive cooling through evapotranspiration

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Technology alone can make cities sustainable"	While technology is necessary — renewable energy, electric vehicles, smart grids, water recycling — technology without systemic redesign merely makes an unsustainable system slightly less harmful. An electric car still requires roads, parking, tire production, and battery materials. A smart building in a car-dependent suburb still generates sprawl. Sustainability requires changing the systems within which technology operates: land use patterns, transportation networks, resource flows, and governance structures.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Dense cities are bad for the environment because they are crowded and polluted"	Dense cities actually have lower per-capita emissions than suburbs and rural areas because they enable public transit, shared walls (less heating/cooling energy per person), shorter distances to services, and more efficient infrastructure. New York City has one-third the per-capita carbon footprint of the average American, primarily because most residents walk, bike, or take transit. The pollution associated with dense cities comes from cars and fossil fuel power plants, not from density itself.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Sustainability means sacrificing quality of life"	The most sustainable cities in the world — Copenhagen, Amsterdam, Vienna, Tokyo — consistently rank among the highest in quality of life. Walkable neighborhoods with good transit, clean air, green spaces, and local services are more pleasant to live in than car-dependent sprawl with long commutes, traffic congestion, and air pollution. Sustainability improvements often increase quality of life while reducing costs: transit is cheaper than car ownership, bike commuting improves health, and green spaces reduce stress.